

KUSHAL KOTHARI

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EDUCATION

New York University — M.S. in Computer Science

Sep 2023 — May 2025

New York, USA

University of Mumbai — B.E. in Computer Engineering

Jul 2018 — Jun 2022

Mumbai, India

SKILLS

Languages: C++, Python, Java, SQL, TypeScript/JavaScript, Erlang

Backend & Distributed Systems: REST APIs (FastAPI, Flask, Django), gRPC, Kafka, Microservices, Asynchronous Processing, Caching (Redis), Multithreading, CI/CD, Docker, Kubernetes, Linux

Cloud (AWS): EC2, S3, Lambda, DynamoDB, SQS, CloudWatch, IAM, Auto Scaling, ALB — AWS Certified Solutions Architect — Associate (2025)

Data & ML: BigQuery, Snowflake, Apache Spark, Airflow, dbt, PostgreSQL, MongoDB, LLM systems (RAG, LangChain), PyTorch

PROFESSIONAL EXPERIENCE

Data Engineer, Optimum Media (Altice USA)

Oct 2025 — Present

New York

- Ship production SQL/DDL data pipelines to a shared WBD repo (feature branches, code review, merge workflow) processing 1B+ household-level TV viewership records across 10+ networks; built the NRP hourly program-alignment view and ratings-enhancement logic now running in production.
- Migrated ratings pipeline logic from GCP/BigQuery to Databricks and Snowflake, resolving broadcast-calendar edge cases and schema differences across systems; validated every output against Databricks as ground truth before client delivery.
- Traced a bronze-to-silver streaming pipeline end-to-end to resolve a measurement discrepancy, proving impressions were captured correctly and quantifying a ~100K/day device-impression attribution gap that informed a business decision on FAST-channel measurability.

Software Engineer, Media.Net

Sep 2022 — Aug 2023

Mumbai

- Automated migration of 1,500+ AdTech client websites to AWS (EC2, S3) with Python orchestration scripts, eliminating manual deployment overhead and cutting launch-critical errors by 20%.
- Configured Auto Scaling Groups and Application Load Balancers with automated health checks and failure detection for high-traffic workloads, saving \$32K in recovery costs.
- Re-architected legacy cron jobs as event-driven AWS Lambda functions, improving runtime performance by 25% and reducing monthly cloud spend.

Open Source Software Engineer, Apache APISIX, Apache Software Foundation

May 2021 — Aug 2021

Remote

- Shipped Fedora RPM packaging for the APISIX API gateway; diagnosed silent post-install failures by tracing missing runtime dependencies (OpenResty, LuaJIT) and hardcoded CentOS paths in the .spec, then refactored packaging Dockerfiles with conditional build args to eliminate ~40% duplicated build logic.
- Built a runtime validation harness (install → service start → HTTP health check) and integrated it into GitHub Actions matrix CI across Fedora and CentOS, cutting clean-install failure rate from 6/10 to 0/10; maintainers adopted the validator as the baseline for new distros, and it caught 2 unrelated regressions in later PRs.

Open Source Developer, Google Summer of Code, Erlang Ecosystem Foundation

Jun 2020 — Aug 2020

Remote

- Integrated 18 Yggdrasil mesh-network admin APIs into the BarrelDB distributed database over UNIX sockets in Erlang; consolidated per-endpoint socket logic into a single shared handler with timeouts and exponential-backoff retries, removing 200+ lines of duplicated code — merged into the main BarrelDB repository.
- Caught and fixed a silent partial-read failure that would have broken replication in production; hardened data synchronization under high node churn and passed all three Google-conducted evaluations.

PROJECTS

Chronicle — Chrome Extension (TypeScript, Vue 3, Manifest V3) *Chrome Web Store · Medium write-up*

- Built and shipped a production browser extension end-to-end: service-worker background processing, cross-session state persistence, and MV3 lifecycle handling; authored a technical deep-dive published on Medium.

HTTP Web Server from Scratch (Python, Sockets, TCP) *GitHub*

- Built an HTTP/1.1 server on raw TCP sockets with no web framework: request parsing, GET and POST handling, static file serving, and correct status-code responses (200/404).
- Implemented persistent keep-alive connections and structured request logging on top of the base server; wrote the socket lifecycle and HTTP response formatting by hand to understand the protocol layer beneath frameworks like Flask.

OPEN SOURCE & PUBLICATIONS

- Linux kernel patch contributor; Open Source Day mentor, Grace Hopper Celebration 2021 (AnitaB.org); NYU Graduate Ambassador.
- Tor vs Yggdrasil: A Comparative Study, IEEE ICICT 2022 — global RTT/packet-loss benchmark of two anonymity networks. IEEE Xplore | Real-Time Object Detection using SSD and MobileNet, IJRASET 2022. Paper